Circuit Documentation

Summary

This circuit is designed to control a robotic car using an ESP32 microcontroller and an L298N motor driver. The ESP32 is programmed to create a Wi-Fi access point, allowing remote control of the car via a web interface. The L298N motor driver is used to control four DC motors, enabling forward, backward, left, and right movements. The circuit is powered by a 12V battery.

Component List

1. **ESP32-WROOM-32 Dev Board**
   - **Description**: A powerful microcontroller with Wi-Fi and Bluetooth capabilities.
   - **Purpose**: Acts as the main controller for the robotic car, handling Wi-Fi communication and motor control signals.
2. **L298N Motor Driver**
   - **Description**: A dual H-bridge motor driver that can control the speed and direction of two DC motors.
   - **Purpose**: Drives the four DC motors of the car, allowing for bidirectional control.
3. **DC Motors (4x)**
   - **Description**: Standard DC motors suitable for small robotic applications.
   - **Purpose**: Provide movement for the robotic car.
4. **12V Battery**
   - **Description**: A power source for the motor driver and motors.
   - **Purpose**: Supplies power to the L298N motor driver and indirectly to the motors.

Wiring Details

ESP32-WROOM-32 Dev Board

- **D14**: Connected to L298N Motor Driver IN1
- **D27**: Connected to L298N Motor Driver IN2
- **D25**: Connected to L298N Motor Driver IN3
- **D33**: Connected to L298N Motor Driver IN4
- **D26**: Connected to L298N Motor Driver ENA
- **VIN**: Connected to L298N Motor Driver 5V
- **GND**: Connected to the common ground with the battery and L298N Motor Driver

L298N Motor Driver

- **12V**: Connected to the positive terminal of the 12V Battery
- **GND**: Connected to the common ground with the ESP32 and battery
- **OUT1**: Connected to the VCC of two DC Motors
- **OUT2**: Connected to the GND of the same two DC Motors

- **OUT3**: Connected to the VCC of the other two DC Motors
- **OUT4**: Connected to the GND of the same two DC Motors
- **IN1**: Connected to ESP32 D14
- **IN2**: Connected to ESP32 D27
- **IN3**: Connected to ESP32 D25
- **IN4**: Connected to ESP32 D33
- **ENA**: Connected to ESP32 D26

DC Motors

- **Motor 1 & 2**:
  - **VCC**: Connected to L298N OUT1
  - **GND**: Connected to L298N OUT2
- **Motor 3 & 4**:
  - **VCC**: Connected to L298N OUT3
  - **GND**: Connected to L298N OUT4

12V Battery

- **Positive (+)**: Connected to L298N 12V
- **Negative (-)**: Connected to the common ground with the ESP32 and L298N Motor Driver

Code Documentation

The ESP32 is programmed using the Arduino framework. The code sets up a Wi-Fi access point and a web server to control the car's movements. The following key functionalities are implemented:

- **Wi-Fi Setup**: The ESP32 creates a Wi-Fi access point named "ESP32_ROBOT" with the password "12345678".
- **Web Server**: A web server is set up to handle HTTP requests for controlling the car.
- **Motor Control**: The ESP32 controls the L298N motor driver using PWM signals to adjust motor speed and direction.
- **HTML Interface**: A simple web interface is provided for user interaction, allowing control of the car's direction and speed.

Key Code Sections

- **Setup Function**: Initializes serial communication, configures GPIO pins, sets up PWM for motor control, and starts the Wi-Fi access point and web server.
- **Loop Function**: Continuously handles incoming client requests.
- **Motor Functions**: Functions like forward(), backward(), left(), right(), and stopCar() control the motor states based on user input.
- **HTML Interface**: Provides a user-friendly interface for controlling the car via a web browser.

This documentation provides a comprehensive overview of the circuit design, component connections, and embedded code functionality for the robotic car project.